

Tutorial 2

1. A wave in water 100 m deep has a period of 10 s and a height of 2 m. Determine the wave celerity, length, and steepness. What is the water particle speed at the wave crest?

Solution:

Assume that this is a deep water wave. Then,

$$L_o = \frac{9.81(10)^2}{2\pi} = 156 \text{ m}$$

Since the depth is greater than half of the calculated wave length, the wave is in deep water and the wave length is 156 m. The wave celerity

$$C_o = \frac{156}{10} = 15.6 \text{ m/s}$$

and the steepness is

$$\frac{H_o}{L_o} = \frac{2}{156} = 0.013$$

For deep water the particle orbits are circular having a diameter at the surface equal to the wave height. Since a particle completes one orbit in one wave period, the particle speed at the crest would be the orbit circumference divided by the period or

$$u_c = \frac{\pi H_o}{T} = \frac{3.14(2)}{10} = 0.63 \text{ m/s}$$

Note that this is much less than C_o .

2. A wave in water 100 m deep has a period of 10 s and a height of 2 m has propagated in to a nearshore depth of 2.3 m. Calculate the wave celerity and length.

Solution:

Assuming this is a shallow water wave, yields

$$C = \sqrt{9.81(2.3)} = 4.75 \text{ m/s}$$

and

$$L = 4.75(10) = 47.5 \text{ m}$$

So $d/L = 2.3/47.5 = 0.048 < 0.05$ and the assumption of shallow water was correct.

3. A wave of height $H = 3$ m and wave period $T = 10$ s, propagates in a water depth of 12 m, the corresponding deep-water wave height $H_o = 3.5$ m. Estimate,

a.

The horizontal and vertical displacements from its mean position at $z = 0$ and $z = -d$.

b.

The maximum water particle displacements at a depth of 7.50 m below SWL, where the wave is in deep water

c.

For the above deep water conditions, show that the water particle displacements are small, relative to the wave height at $z = L_o/2$.

d.

Also, compare the water particle displacements in deep water conditions for the corresponding deep-water wave height, $H_o = 3.5$ m and wave period $T = 15$ s at $z = L_o/2$ and $z = -7.5$ m.

d/L_0	d/L	$2\pi d/L$	$\tanh$ ($2\pi d/L$)	$\sinh$ ($2\pi d/L$)	$\cosh$ ($2\pi d/L$)	H/H'_0	K	$4\pi d/L$	$\sinh$ ($4\pi d/L$)	$\cosh$ ($4\pi d/L$)	n	C_g/C_0	M
0.0710	0.1149	0.7218	0.6180	0.7861	1.2720	0.9694	0.7862	1.4436	2.0000	2.2361	0.8609	0.5321	12.92
0.0720	0.1158	0.7277	0.6217	0.7937	1.2767	0.9676	0.7833	1.4554	2.0265	2.2598	0.8591	0.5341	12.77
0.0730	0.1168	0.7336	0.6253	0.8012	1.2814	0.9658	0.7804	1.4672	2.0532	2.2837	0.8573	0.5360	12.62
0.0740	0.1177	0.7394	0.6288	0.8087	1.2861	0.9641	0.7776	1.4789	2.0800	2.3079	0.8555	0.5379	12.48
0.0750	0.1186	0.7453	0.6323	0.8162	1.2908	0.9624	0.7747	1.4905	2.1071	2.3323	0.8537	0.5398	12.34
0.0760	0.1195	0.7511	0.6358	0.8237	1.2956	0.9608	0.7719	1.5021	2.1343	2.3570	0.8519	0.5416	12.21
0.0770	0.1205	0.7569	0.6392	0.8312	1.3004	0.9592	0.7690	1.5137	2.1618	2.3819	0.8501	0.5434	12.08
0.0780	0.1214	0.7626	0.6426	0.8387	1.3052	0.9577	0.7662	1.5253	2.1894	2.4070	0.8483	0.5452	11.95
0.0790	0.1223	0.7684	0.6460	0.8463	1.3100	0.9562	0.7633	1.5368	2.2173	2.4323	0.8466	0.5469	11.83
0.0800	0.1232	0.7741	0.6493	0.8538	1.3149	0.9547	0.7605	1.5483	2.2453	2.4579	0.8448	0.5485	11.70
0.0810	0.1241	0.7798	0.6526	0.8613	1.3198	0.9533	0.7577	1.5597	2.2736	2.4838	0.8430	0.5502	11.59
0.0820	0.1250	0.7855	0.6559	0.8689	1.3247	0.9519	0.7549	1.5711	2.3020	2.5099	0.8412	0.5518	11.47
0.0830	0.1259	0.7912	0.6591	0.8764	1.3297	0.9506	0.7521	1.5825	2.3307	2.5362	0.8395	0.5533	11.36
0.0840	0.1268	0.7969	0.6623	0.8840	1.3347	0.9493	0.7492	1.5938	2.3596	2.5628	0.8377	0.5548	11.25
0.0850	0.1277	0.8026	0.6655	0.8915	1.3397	0.9480	0.7464	1.6051	2.3888	2.5896	0.8360	0.5563	11.14
0.0860	0.1286	0.8082	0.6686	0.8991	1.3448	0.9468	0.7436	1.6164	2.4181	2.6168	0.8342	0.5578	11.04
0.0870	0.1295	0.8138	0.6717	0.9067	1.3498	0.9456	0.7408	1.6276	2.4477	2.6441	0.8325	0.5592	10.94
0.0880	0.1304	0.8194	0.6748	0.9143	1.3550	0.9444	0.7380	1.6389	2.4776	2.6718	0.8307	0.5606	10.84
0.0890	0.1313	0.8250	0.6778	0.9219	1.3601	0.9433	0.7352	1.6501	2.5076	2.6997	0.8290	0.5619	10.74
0.0900	0.1322	0.8306	0.6808	0.9295	1.3653	0.9422	0.7325	1.6612	2.5379	2.7278	0.8273	0.5632	10.65

Solution

Given data: $H = 3$ m, $T = 10$ s and $H_o = 3.5$ m

The maximum horizontal (D) and vertical (B) displacements of water particle.

$L_o = 1.56T^2 = 1.56(10)^2 = 156$ m, $d/L_o = 12/156 = 0.07692$ From wave tables of SPM (1984), $d/L = 0.1205$ and hence $L = 99.58$ m $kd = 2\pi/L = 0.0631$

a.

At $z = 0$

$$D = \frac{H \cos hk(d+z)}{2 \sin hkd}, B = \frac{H \sin hk(d+z)}{2 \sin hkd}$$

Substituting for the parameters, $D = 2.34$ m and $B = 1.5$ m

At $z = -12$ m

$$D = \frac{H}{2} \frac{1}{\sin hkd} = \frac{3}{2} \left(\frac{1}{0.8316} \right) = 1.80 \text{ m and } B = 0$$

b.

In deep water conditions, at $z = -7.5$ m

$$k_o = \frac{2\pi}{L_o} = \frac{2\pi}{156} = 0.0402 \quad D = B = \frac{H_o}{2} e^{k_o z} = \frac{3.5}{2} e^{(0.0402 \times (-7.5))} = 1.2938 \text{ m}$$

c.

In deep water conditions, at $z = -L_o/2$

$$z = -156/2 = -78 \text{ m}$$

$$\text{As } H_o = 3.5 \text{ m, } D = B = \frac{H_o}{2} e^{k_o z} = \frac{3.5}{2} e^{[0.0402 \times (-78)]} = 0.07562 \text{ m.}$$

It is inferred that for deep-water conditions, the particle displacements are very small at $z = -7.5$ m compared to at $z = L_o/2$ by a factor of $1.2938/0.07562 = 17.1$.

d.

$$T = 15 \text{ s}, H_o = 3.5 \text{ m}.$$

$$L_o = 1.56T^2 = 1.56(15)^2 = 351 \text{ m}, k_o = 2\pi/L_o = 0.0179 \text{ At } z = -L_o/2 = -351/2 = -175.5 \text{ m}$$

$$D = B = \frac{H_o}{2} e^{k_o z} = \frac{3.5}{2} e^{(0.0179 \times (-175.5))} = 0.07563 \text{ m}.$$

$$\text{At } z = -7.5 \text{ m}$$

$$D = B = \frac{H_o}{2} e^{k_o z} = \frac{3.5}{2} e^{(0.0179 \times (-7.5))} = 1.53 \text{ m}$$

$$(\text{Displacement at } z = -7.5 \text{ m}) / (\text{displacement at } z = 175.5 \text{ m}) = 1.53/0.07563 = 20.2$$